

LABORATORY OBSERVATION/RECORD

Course: Full Stack Web Development
Course Code: 23CM4121
Experiment No.: 1
Student Name: K. Sirisha
Roll No.: A24126552260
Section: B

1. TITLE

Design a College Information Portal Webpage

2. AIM

To design a multi-section College Information Portal webpage using HTML5 and CSS3 that presents information about the college, its departments and contact details.

3. OBJECTIVE

The objectives of this experiment are:

- To understand the basic structure of an HTML5 document.
- To apply text formatting tags and organize content using headings, paragraphs and lists.
- To create hyperlinks, tables and insert images with alternate text.
- To design a student registration form using various form controls.
- To use HTML5 semantic elements for a well-structured layout.
- To embed audio and video using HTML5 media elements.
- To use HTML5 input types such as Email, Date, Number, Color and Range.

4. THEORY

HTML5 is the latest version of the Hyper Text Markup Language used to structure content on the web. It introduces semantic elements such as `<header>`, `<nav>`, `<section>`, `<article>` and `<footer>` that describe the meaning of the content they contain, making pages easier to read and maintain.

Text formatting tags such as `<b>`, `<i>`, `<u>`, `<sup>` and `<sub>` are used to change how text is displayed. Lists are created using `<ol>` for ordered lists and `<ul>` for unordered lists. The `<table>` element, along with `<tr>`, `<th>` and `<td>`, is used to display tabular data such as student details.

HTML5 forms use elements like `<input>`, `<select>` and `<textarea>` to collect user data. New input types introduced in HTML5 — email, date, number, color and range — provide built-in validation and specialized controls in the browser. HTML5 also introduced native `<audio>` and `<video>` elements to embed media without external plugins.

5. ALGORITHM / PROCEDURE

1. Create a new HTML file and declare the HTML5 document type.
2. Add the `<head>` section with a title and internal CSS styling.
3. Build the page structure using semantic tags — header, nav, section, article and footer.
4. Add headings and paragraphs describing the college, using text formatting tags.
5. Create one ordered list (departments) and one unordered list (facilities).
6. Add hyperlinks to the college website and to Google.
7. Insert the college logo and a relevant campus image with alternate text.
8. Create a table displaying student Roll No, Name, Branch and Email.
9. Design a registration form with text, email, password, radio, checkbox, dropdown and textarea fields.
10. Add HTML5 input types — Date, Number, Color and Range — inside the form.
11. Embed an audio player and a video player using the audio and video elements.
12. Save the file and open it in a web browser.
13. Verify that every section renders and functions as expected.

6. PROGRAM

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<title>College Information Portal</title>
<style>
  body { font-family: Arial, sans-serif; margin: 20px; color: #222; }
  header, nav, section, article, footer { display: block; }
  header { background: #1a3c6e; color: #fff; padding: 15px 20px; }
  nav { background: #14305a; padding: 10px 20px; }
  nav a { color: #fff; margin-right: 18px; text-decoration: none; font-weight: bold; }
  section { padding: 15px 20px; }
  h1, h2 { margin-top: 0; }
  table { border-collapse: collapse; width: 60%; margin: 10px 0; }
  table, th, td { border: 1px solid #444; }
  th, td { padding: 8px 12px; text-align: left; }
  th { background: #dce6f2; }
  form { max-width: 480px; }
  form label { display: inline-block; width: 110px; margin-top: 8px; }
  fieldset { margin-top: 10px; }
  footer { background: #1a3c6e; color: #fff; padding: 12px 20px; text-align: center; margin-top: 20px; }
  .img-row img { margin-right: 20px; vertical-align: top; }
</style>
</head>
<body>

<header>
  <h1>College Information Portal</h1>
</header>

<nav>
  <a href="#about">About</a>
  <a href="#students">Students</a>
  <a href="#register">Registration</a>
  <a href="#media">Media</a>
</nav>

<section id="about">
  <article>
    <h2>About the College</h2>
    <p><b>ABC Institute of Technology</b> is a premier engineering institution established in 1998, offering undergraduate and postgraduate programs in <i>Computer Science, Electronics, and Mechanical Engineering</i>. The college is <u>NAAC 'A' accredited</u> and is affiliated to the State Technical University.</p>

    <p>Our chemistry lab works with compounds such as H<sub>2</sub>O and CO<sub>2</sub>, and our physics lab explores concepts like  $E = mc^2$ .</p>

    <h3>Ordered List – Departments</h3>
    <ol>
      <li>Computer Science and Engineering</li>
      <li>Electronics and Communication Engineering</li>
      <li>Mechanical Engineering</li>
      <li>Civil Engineering</li>
    </ol>

    <h3>Unordered List – Facilities</h3>
    <ul>
      <li>Central Library</li>
      <li>Sports Complex</li>
      <li>Hostel Accommodation</li>
      <li>Wi-Fi Enabled Campus</li>
    </ul>

    <p>Visit our <a href="https://www.example-college.edu" target="_blank">College Website</a> or search more on <a href="https://www.google.com" target="_blank">Google</a>.</p>
  </article>
</section>

<section id="images">
  <h2>Gallery</h2>
  <div class="img-row">
    
    
  </div>
</section>

<section id="students">
  <h2>Student Details</h2>
  <table>
    <tr><th>Roll No</th><th>Name</th><th>Branch</th><th>Email</th></tr>
    <tr><td>23CM4101</td><td>P. Santosh Kumar</td><td>CSE</td><td>santosh@example.edu</td></tr>
    <tr><td>23CM4102</td><td>K. Priya</td><td>ECE</td><td>priya@example.edu</td></tr>
    <tr><td>23CM4103</td><td>M. Arjun</td><td>MECH</td><td>arjun@example.edu</td></tr>
  </table>
</section>
```

```

</table>
</section>

<section id="register">
  <h2>Student Registration Form</h2>
  <form action="#" method="POST">
    <label for="name">Name:</label>
    <input type="text" id="name" name="name" required><br><br>

    <label for="email">Email:</label>
    <input type="email" id="email" name="email" required><br><br>

    <label for="password">Password:</label>
    <input type="password" id="password" name="password" required><br><br>

    <label>Gender:</label>
    <input type="radio" id="male" name="gender" value="male"><label for="male" style="width:auto">Male</label>
    <input type="radio" id="female" name="gender" value="female"><label for="female" style="width:auto">Female</label><br><br>

    <label>Hobbies:</label>
    <input type="checkbox" id="reading" name="hobbies" value="reading"><label for="reading" style="width:auto">Reading</label>
    <input type="checkbox" id="sports" name="hobbies" value="sports"><label for="sports" style="width:auto">Sports</label>
    <input type="checkbox" id="music" name="hobbies" value="music"><label for="music" style="width:auto">Music</label><br><br>

    <label for="branch">Branch:</label>
    <select id="branch" name="branch">
      <option value="cse">CSE</option>
      <option value="ece">ECE</option>
      <option value="mech">MECH</option>
      <option value="civil">CIVIL</option>
    </select><br><br>

    <label for="dob">Date of Birth:</label>
    <input type="date" id="dob" name="dob"><br><br>

    <label for="age">Age:</label>
    <input type="number" id="age" name="age" min="16" max="60"><br><br>

    <label for="color">Favourite Color:</label>
    <input type="color" id="color" name="color"><br><br>

    <label for="marks">Marks Range:</label>
    <input type="range" id="marks" name="marks" min="0" max="100"><br><br>

    <label for="comments">Comments:</label><br>
    <textarea id="comments" name="comments" rows="4" cols="40"></textarea><br><br>

    <button type="submit">Submit</button>
    <button type="reset">Reset</button>
  </form>
</section>

<section id="media">
  <h2>Media</h2>
  <p>Audio Player:</p>
  <audio controls>
    <source src="sample-audio.mp3" type="audio/mpeg">
    Your browser does not support the audio element.
  </audio>

  <p>Video Player:</p>
  <video controls width="320">
    <source src="sample-video.mp4" type="video/mp4">
    Your browser does not support the video element.
  </video>
</section>

<footer>
  &copy; 2026 ABC Institute of Technology – College Information Portal
</footer>

</body>
</html>

```

7. CODE EXPLANATION

Code	Explanation
<header>, <nav>, <section>, <article>, <footer>	HTML5 semantic elements used to structure the page into meaningful regions.
, <i>, <u>, <sup>, <sub>	Format text as bold, italic, underlined, superscript and subscript.
/	Display the department list (ordered) and facilities list (unordered).
	Creates hyperlinks to the college website and Google.
	Displays the college logo and campus image; alt text shows if the image fails to load.
<table>, <tr>, <th>, <td>	Builds the student details table with Roll No, Name, Branch and Email.
<form>, <input>, <select>, <textarea>	Collects student registration data through various form controls.
type="radio" / "checkbox"	Lets the user select Gender (one option) and Hobbies (multiple options).
type="date" / "number" / "color" / "range"	HTML5 input types providing built-in pickers and validation in the browser.
<audio controls> / <video controls>	Embeds a playable audio and video element with default browser controls.

8. TEST CASES AND EXECUTION DATA

The webpage was opened in a browser and each section was verified against its expected behaviour.

Test Case	Action / Input	Expected Result	Actual Result	Status
TC-01	Load the page in browser	Header, nav bar, about section and lists display correctly	Displayed correctly	Pass
TC-02	Click "College Website" and "Google" links	Links open the respective pages in a new tab	Opened correctly	Pass
TC-03	Check logo and campus image	Both images render with correct alternate text	Rendered correctly	Pass
TC-04	Check student details table	Roll No, Name, Branch and Email shown in a grid	Displayed correctly	Pass
TC-05	Submit registration form with Name field empty	Browser should require the Name field	Browser blocked submission and asked to fill Name	Pass
TC-06	Select Gender and one or more Hobbies	Only one Gender selectable; multiple Hobbies selectable	Behaved correctly	Pass
TC-07	Pick a Date, Number, Color and Range value	Native browser pickers open for each input type	Pickers opened as expected	Pass
TC-08	Play the Audio and Video elements	Media controls (play/pause) are visible and functional	Controls displayed and functional	Pass

9. OUTPUT

Output 1: Header, About Section, Text Formatting, Lists, Hyperlinks & Gallery

College Information Portal

About Students Registration Media

About the College

ABC Institute of Technology is a premier engineering institution established in 1998, offering undergraduate and postgraduate programs in *Computer Science, Electronics, and Mechanical Engineering*. The college is NAAC 'A' accredited and is affiliated to the State Technical University.

Our chemistry lab works with compounds such as H_2O and CO_2 , and our physics lab explores concepts like $E = mc^2$.

Ordered List — Departments

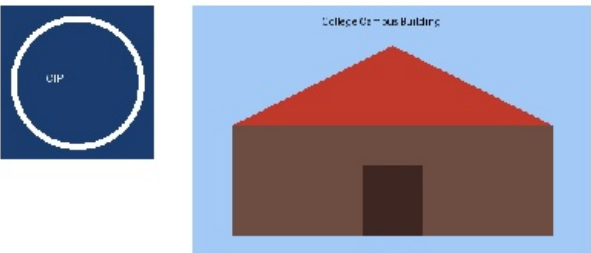
1. Computer Science and Engineering
2. Electronics and Communication Engineering
3. Mechanical Engineering
4. Civil Engineering

Unordered List — Facilities

- Central Library
- Sports Complex
- Hostel Accommodation
- Wi-Fi Enabled Campus

Visit our [College Website](#) or search more on [Google](#).

Gallery



Student Details

Roll No	Name	Branch	Email
23CM4101	P. Santosh Kumar	CSE	santosh@example.edu
23CM4102	K. Priya	ECE	priya@example.edu
23CM4103	M. Arjun	MECH	arjun@example.edu

Student Registration Form

Name:

Output 2: Student Table & Registration Form

Name:

Email:

Password:

Gender:

Male

Female

Hobbies:

Reading

Sports

Music

Branch:

CSE

Date of Birth:

Age:

Favourite

Student details table and the top of the registration form (Name, Email, Password, Gender, Hobbies, Branch dropdown).

Output 3: HTML5 Input Types, Media Elements & Footer

Favourite
Color:



Marks Range:

Comments:

Submit

Reset

Media

Audio Player:



Video Player:



© 2026 ABC Institute of Technology — College Information Portal

Date, Color and Range input types, Submit/Reset buttons, HTML5 audio and video players with controls, and the semantic footer.

10. RESULT

Thus, a College Information Portal webpage was successfully designed using **HTML5 and CSS3**. Text formatting, lists, hyperlinks, tables, images, a registration form, semantic elements, HTML5 input types and embedded media were implemented, and the output was verified against the expected results for all test cases.